

FT-Behavioural Router: Knowledge vs Behaviour Separation in Adaptive Hybrid Inference

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Abstract

Hybrid assistants combine retrieval-augmented generation (RAG), tools, and multiple model sizes, but calling a large LLM with retrieval on every query is costly. We study a fine-tuned (FT) behavioural router that selects among four actions: answer as a small model, retrieve and answer as the same small model (FT+RAG), invoke tools, or escalate to a large LLM, while keeping factual knowledge non-parametric. We do *not* claim inventing adaptive RAG or LLM cascading. On in-domain workplace eval_v2 ($n=133$), FT DistilBERT reaches accuracy 0.9699 and MiniLM 0.8872, above a prompt-rubric simulation (0.7744; **not** an LLM API). Exact McNemar and paired bootstrap tests on the same eval items find DistilBERT significantly above MiniLM and keyword (measured $p<0.01$; artefact results/acl/significance_*.json). On a hard OOD split ($n=104$; never used for training) DistilBERT drops to 0.8750 and the rubric collapses to 0.5000. A remapped public eval (SQuAD 1.1, HotpotQA, CLINC-150; $n=240$; deterministic labels, not IAA) is a second domain: workplace cue rules fall to ≈ 0.25 – 0.28 ; workplace-trained DistilBERT transfers at 0.5000 while remapped-train DistilBERT reaches 0.9250 (source-style caveat). A Phase 2 unit-cost simulation shows FT mean cost roughly $0.25\times$ always-RAG+escalate; this is **not** a cloud bill and no live answer EM/F1 is reported. We deepen the related-work contrast, error analysis, limitations, and when-FT-is-worth-it criteria for an *ACL Findings-oriented cut.

1 Introduction

Modern assistant stacks mix RAG (Lewis et al., 2020), tool calling, and several model sizes. Calling a large LLM with retrieval on every query is expensive; answering everything from a small parametric model fails on freshness and hard reasoning. The decision of *which capability to invoke* therefore

controls cost, latency, and reliability.

Prior work covers pieces of this decision: when to retrieve (Jeong et al., 2024; Tan et al., 2024; Asai et al., 2023; Yan et al., 2024; Yu et al., 2024), which input augmentation to apply for one LLM (Chen et al., 2025), and which model size to call (Ong et al., 2025; Ding et al., 2024; Chen et al., 2023; Madaan et al., 2024; Shen et al., 2025). What is less studied is a unified policy that jointly chooses among answering as a small model, answering as the same small model with retrieval (FT+RAG), calling tools, and escalating to a large LLM, with facts kept outside the fine-tuned weights.

This paper reports two phases. Phase 1 is *measured* 4-way routing classification on in-domain, hard out-of-distribution (OOD), and remapped public splits, with paired significance tests on eval_v2. Phase 2 is a *simulation* under a fixed unit cost model (not live API bills, not live answer quality).

Non-claims. Adaptive retrieval and cascading are not claimed as new. The prompt-rubric baseline is not an LLM-API router (a live prompted path exists behind API keys; offline runs are labelled as such). High in-domain DistilBERT accuracy is not read as near-solved routing; OOD and transfer probes are reported for that reason. Workplace labels are author policy until a human double-annotation audit completes (protocol and blank second-annotator spreadsheet are in the repository).

Contributions. (1) A unified 4-way behavioural policy, where `rag` means FT+RAG (small model + retrieved docs) and `large+RAG` is `escalate_large` with `needs_rag`. (2) An explicit knowledge $\neq$ behaviour split: facts stay in the corpus/APIs; only the routing policy is fine-tuned. (3) Measured when-is-FT-worth-it criteria versus majority, keyword, and prompt-rubric baselines, with in-domain, OOD, remapped public

transfer, bootstrap, k -fold, and McNemar / paired-bootstrap numbers.

2 Related Work

Adaptive retrieval. Adaptive-RAG (Jeong et al., 2024) trains a classifier to route no-retrieval / single-step / multi-step RAG by query complexity. SlimPLM (Tan et al., 2024) uses a slim proxy to judge when and what to retrieve. Self-RAG (Asai et al., 2023) embeds retrieve/critique decisions inside one generator via reflection tokens. CRAG (Yan et al., 2024) and Auto-RAG (Yu et al., 2024) improve or iterate retrieval quality; they do not define a behavioural controller over tools and escalate-to-large as peer actions.

Query augmentation routing. QueryRouter (Chen et al., 2025) routes *input augmentation* (CoT, rewrite, retrieval, none) for a single LLM. It is a close neighbour on fine-tuned lightweight routing, but the decision space is input augmentation for one model, not a hybrid controller where answering as a small model, calling tools, or escalating are peer actions.

Model cascading. RouteLLM (Ong et al., 2025), HybridLLM (Ding et al., 2024), FrugalGPT (Chen et al., 2023), AutoMix (Madaan et al., 2024), and SATER (Shen et al., 2025) study strong-weak or cascade selection. These typically omit RAG and tools from the same primary action space.

RAG-aware selection. RAGRouter (Zhang et al., 2025) selects among multiple RAG-enabled LLMs; UniversalRAG (Yeo et al., 2025) routes across corpora/modalities. They choose *which knowledge path or which model under RAG*, not the 4-way behavioural space studied here.

Gap. Prior work covers (a) when to retrieve, (b) which augmentation for one LLM, (c) which model size, and (d) which corpus/model under RAG (Table 1). What remains thin is a single behavioural policy over $\{\text{answer_small}, \text{rag}, \text{tools}, \text{escalate_large}\}$ with $\text{knowledge} \neq \text{behaviour}$ training and a clear FT-vs-prompt worth-it study under shift. FT-last rewriting of a finished LLM answer is not the research core.

3 Method

3.1 Action space and execution paths

The router R_θ maps a query q to a primary action $a \in \{\text{answer_small}, \text{rag}, \text{tools}, \text{escalate_large}\}$.

Table 2 shows five execution paths from four labels plus `needs_rag`. `rag` is FT+RAG: retrieve company or time-sensitive facts, then the *same* small FT model writes the answer. Facts are not written into FT weights, so the corpus can be updated without retraining the behavioural model.

Design principles. (1) **Knowledge $\neq$ behaviour:** the small model is not fine-tuned to memorise domain facts. (2) **Small may answer alone or with RAG.** (3) **Large owns hard reasoning.** (4) **Freshness via RAG/tools.**

3.2 Data

Workplace-style queries are labelled with a mutually exclusive primary action (schema 1.0; decision order: `tools` $\rightarrow$ `rag` $\rightarrow$ `escalate` $\rightarrow$ `answer_small`). Splits: `train_v2` $n=526$, `eval_v2` $n=133$, `eval_ood` $n=104$ (held-out hardness probe; never used for training or selection). A second domain, `eval_public_remap` ($n=240$), remaps public SQuAD 1.1 (Rajpurkar et al., 2016), HotpotQA (Yang et al., 2018), and CLINC-150 (Larson et al., 2019) into the same schema with deterministic rules (not human IAA).

OOD construction. Hard OOD stresses ambiguity, multi-intent, tools-vs-rag paraphrases, entity-heavy queries, and adversarial keyword traps. If in-domain scores drop sharply on OOD, that is evidence the v2 labels are partly too easy.

Label audit (in progress). A double-annotation protocol and stratified sample ($n=100$; `annotator_b` blank) are provided in the repository. Until a second annotator fills blanks, we report **no** Cohen’s κ and do not invent agreement numbers.

3.3 Models and baselines

Non-neural: majority; keyword regex; a prompt-rubric simulation that scores LABELING.md cues (not an LLM API). A prompted LLM router interface calls OpenAI/Anthropic when keys are present; otherwise the same cue rubric runs offline and results are marked `live_llm=false`.

Neural	FT:	MiniLM
(all-MiniLM-L6-v2)	and	DistilBERT

Table 1: Capability and decision-space contrast (✓ present, ∼ partial, — absent). $K \neq B$ = explicit knowledge $\neq$ behaviour training. Worth-it = reports when a lightweight FT router beats a strong non-neural / prompted baseline under shift.

System	RAG	Tools	Small ans.	Esc. large	$K \neq B$	Decision space	Worth-it	Shift probe
Adaptive-RAG	✓	—	∼	—	—	retrieve complexity	—	QA domains
QueryRouter	✓	—	∼	—	—	input aug. (1 LLM)	∼	efficiency
SlimPLM	✓	—	∼	∼	∼	when/what retrieve	—	QA
Self-RAG / CRAG / Auto-RAG	✓	—	—	—	—	in-LM retrieve	—	QA
RouteLLM / HybridLLM	—	—	✓	✓	—	strong-weak	∼	chat / QA
FrugalGPT / AutoMix / SATER	—	—	✓	✓	—	cascade / verify	∼	API suites
RAGRouter	✓	—	—	✓	—	which RAG-LLM	—	multi-RAG
UniversalRAG	✓	—	—	—	—	corpus / modality	—	multi-modal
This work	✓	✓	✓	✓	✓	4-way behaviour	✓	OOD + public remap

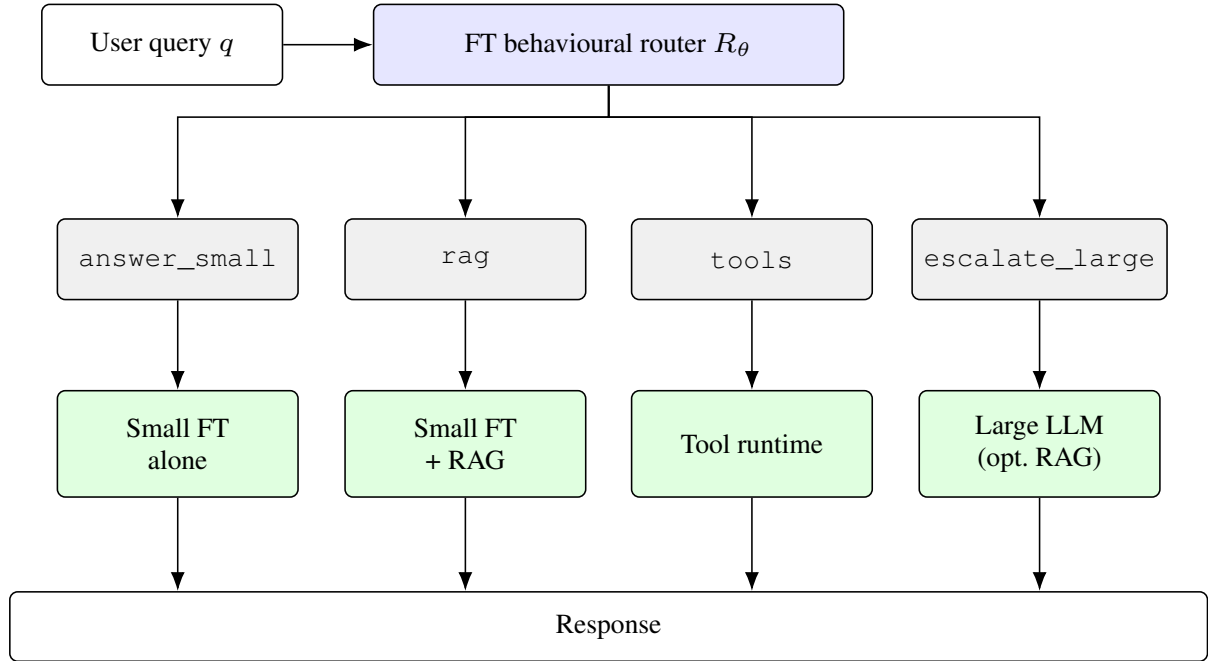


Figure 1: FT-first control. rag is the small FT model plus retrieval. The large LLM is used only on escalate_large.

Table 2: Execution paths from the 4-way labels.

Path	Label	Who answers	Retrieval
FT alone	answer_small	small FT	no
FT+RAG	rag	small FT	yes
Tools	tools	tool runtime	optional
Large alone	escalate_large	large LLM	no
Large+RAG	esc.+needs_rag	large LLM	yes

Table 3: Phase 1 primary-action metrics on eval_v2 ($n=133$). Neural: seed 42; bootstrap mean $\pm$ std, $B=1000$.

Method	Acc.	Macro-F1	Boot. acc.
Majority	0.3083	0.1178	—
Keyword	0.5789	0.5302	—
Prompt-rubric sim.	0.7744	0.7658	—
FT MiniLM	0.8872	0.8829	.886 $\pm$.027
FT DistilBERT	0.9699	0.9692	.969 $\pm$.015

(distilbert-base-uncased), 5 epochs, batch 16, lr 2×10^{-5} , seed 42, max length 128.

Metrics: accuracy, macro-F1, confusion matrices (fixed label order); bootstrap $B=1000$; stratified 5-fold; 24 gold paraphrase pair agreement; exact McNemar and paired bootstrap on per-example correctness.

Phase 2: documented unit cost/latency table ap-

plied to gold and DistilBERT routes, plus local BM25 hit@ k and tool-stub schema checks.

Table 4: Stratified 5-fold on combined train+eval ($n=659$), 3 epochs/fold.

Method	Acc. (mean $\pm$ std)	F1 (mean $\pm$ std)
FT MiniLM	.766 $\pm$.045	.740 $\pm$.051
FT DistilBERT	.910 $\pm$.022	.908 $\pm$.022

Table 5: Exact McNemar and paired bootstrap on eval_v2 ($n=133$, $B=1000$). $\Delta\text{Acc} = \text{Acc}(A) - \text{Acc}(B)$. $b/c = A\text{-correct-}B\text{-wrong} / A\text{-wrong-}B\text{-correct}$. Measured from results/acl/significance_eval_v2.json (seed 42 re-run; DistilBERT acc 0.9699, MiniLM 0.8797 on this run).

A vs B	ΔAcc	b	c	McNemar p
DistilBERT vs MiniLM	+0.0902	14	2	0.0042
DistilBERT vs keyword	+0.3910	55	3	2.26×10^{-13}
MiniLM vs keyword	+0.3008	47	7	2.29×10^{-8}
DistilBERT vs rubric	+0.1955	29	3	2.56×10^{-6}

4 Results

4.1 Phase 1: routing classification

On eval_v2 (Table 3), both FT models beat the strong rubric. k -fold means (Table 4) are lower than single-split point estimates. DistilBERT’s near-ceiling in-domain score is treated as suspicious relative to label simplicity.

4.2 Paired significance (measured)

Table 5 reports paired tests on identical eval items (not unpaired accuracy CIs alone). All p -values are exact McNemar (binomial) on paired correctness; bootstrap CIs are percentile intervals on ΔAcc . DistilBERT significantly exceeds MiniLM, keyword, and the prompt-rubric on this split ($p < 0.01$). MiniLM exceeds keyword ($p < 10^{-7}$) and the rubric ($p \approx 0.024$).

4.3 Hard OOD generalisation

Rule baselines lose the most (rubric $\Delta \approx -0.27$ to 0.50). FT drops ≈ 9 –10 points but stays above the collapsed rubric (Table 6). A 0.97 in-domain score is therefore not near-solved routing.

4.4 Error analysis

In-domain DistilBERT (eval_v2). With $n=133$ and accuracy 0.9699, only four errors remain. The confusion matrix (rows=true, cols=pred; order answer_small, rag, tools, escalate_large) shows: one answer_small $\rightarrow$ tools, two rag $\rightarrow$ tools,

Table 6: In-domain vs hard OOD ($n=104$). FT trained on train_v2 only.

Method	Acc ID	Acc OOD	Δ Acc	F1 OOD
Majority	.308	.260	−.049	.103
Keyword	.579	.365	−.214	.348
Prompt-rubric	.774	.500	−.274	.470
FT MiniLM	.887	.788	−.099	.792
FT DistilBERT	.970	.875	−.095	.874

and one escalate_large $\rightarrow$ tools. All four mistakes land in tools: the operationally expensive direction of over-calling tools on queries that should stay parametric, retrieve-only, or escalate.

There is no in-domain under-escalation of gold escalate_large to answer_small.

MiniLM, in-domain. Errors are more diffuse: rag confuses with answer_small and escalate_large; tools sometimes escalates. Relative to DistilBERT, MiniLM trades a lower ceiling for a less tools-skewed residual.

Hard OOD DistilBERT. OOD confusion shows under-escalation: 5/22 gold escalate_large mis-routed (mostly to tools)—the operationally dangerous direction when hard reasoning is required. rag also leaks into tools (5/31). Under-escalation on OOD is the first-class residual risk for any live deployment, especially for safety-adjacent or multi-hop queries.

Public remap transfer. Workplace $\rightarrow$ public DistilBERT transfer accuracy 0.5000 with **zero** Hotpot escalate_large recall (most Hotpot items collapse to rag). This is consistent with learning workplace lexical cues rather than a domain-general complexity policy.

Paraphrase consistency. On 24 gold paraphrase pairs: DistilBERT 0.917; MiniLM 0.875; rubric 0.917; keyword 0.750. FT does not uniquely own paraphrase stability on this small probe.

4.5 Second domain: public remapped eval

Workplace eval_ood is still author-built company English. Public items are remapped into the same 4-way JSONL: SQuAD extractive $\rightarrow$ rag; HotpotQA bridge/comparison $\rightarrow$ escalate_large+needs_rag; CLINC high-precision intents $\rightarrow$ tools/answer_small. This complexity split is an Adaptive-RAG cousin (Jeong et al., 2024), not a new retriever.

Table 7: Public remapped eval ($n=240$). Remap: train on remapped public. Transfer: train on workplace train_v2 only. Measured: results/acl/.

Method	Acc. remap	F1 remap	Acc. transfer
Majority	.250	.100	.250
Keyword	.254	.117	.254
Prompt-rubric	.283	.168	.283
FT MiniLM	.775	.769	.354
FT DistilBERT	.925	.925	.500

Table 8: Phase 2 simulation (fixed unit costs; **not** a cloud bill).

Policy	Cost ID	Cost OOD	Large ID	Large OOD
Always escalate	1.000	1.000	1.00	1.00
Always RAG+esc.	1.150	1.150	1.00	1.00
FT DistilBERT	0.287	0.256	0.218	0.212
Gold oracle	0.303	0.330	0.226	0.212

Workplace cue rules collapse to near chance (Table 7). Remapped-train DistilBERT is learnable (0.9250) but can use source style. Workplace→public transfer is stricter: DistilBERT 0.5000, MiniLM 0.3542.

4.6 Phase 2: simulated cost

Under the unit model (Table 8), FT DistilBERT mean cost is about $0.25\times$ always-RAG+escalate. On OOD, simulated FT cost (0.256) is *lower* than gold (0.330) because under-escalation routes hard queries to cheaper actions—the simulation flatters FT cost by converting quality loss into apparent savings. Local BM25 hit@1=0.984 (62 primary-RAG qrels); tool-stub schema success 1.000. Neither is live product quality.

4.7 When is FT worth it?

1. **Beat the rubric under paired tests.** If FT macro-F1 does not exceed the prompt-rubric by more than paired uncertainty, ship the rubric (Table 9).
2. **Account for training and drift cost.** A one-day DistilBERT train is cheap; weekly rellabelling is not.
3. **Prefer FT when tools/RAG/escalate confusion is costly.** Over-tools (in-domain) and under-escalate (OOD) are the residual modes to monitor.
4. **Prefer rubric when labels are scarce** or the behavioural policy changes weekly.

Table 9: Decision checklist: when to ship FT vs a strong rubric / keyword policy.

Criterion	Prefer FT	Prefer rubric / keyword
Beat strong non-neural	Macro-F1 exceeds rubric beyond paired uncertainty (McNemar / boot CI)	FT within noise of rubric
Label supply	Stable policy, $\geq$ few hundred labels, drift months	Labels scarce or policy changes weekly
Error cost	Tools/RAG/escalate confusion is expensive (ops / safety)	Mistakes cheap; humans in the loop
	Holds under hard OOD / second domain	Collapses under natural shift
Cost accounting	Training + monitoring budget available; do not equate unit sim with cloud \$/query	Need zero training ops

5. **Do not equate simulation with cloud \$/query.** Phase 2 under-escalation can look like a cost win while being a quality loss.

6. **Do not treat remapped-train accuracy as domain-general.** Transfer 0.5000 is the honest number for workplace-only FT.

Verdict. On this synthetic workplace policy at $n\approx 500/130$, DistilBERT clears the rubric bar under bootstrap, k -fold, hard OOD (0.875 vs rubric 0.50), and paired significance vs keyword/MiniLM. Absolute 0.97 in-domain must not be oversold: report OOD 0.875 alongside it. On the public remap, FT is worth it *after* remapped labels exist; workplace FT alone is not a general 4-way policy (transfer 0.5000).

Limitations

Workplace labels are synthetic constructions, not production logs; they encode an author decision policy (LABELING.md), so in-domain accuracy partly measures fit to that policy. The public second domain uses *deterministic remaps*, not human gold; inter-annotator agreement is not yet measured (a double-annotation audit protocol and blank second-annotator spreadsheet are started in the repository; κ remains pending). Phase 1 is routing-label accuracy only—not answer EM/F1, tool-execution success, or user preference. Phase 2 is a fixed-unit simulation: no live LLM API bills and no

answer quality; under-escalation can flatter simulated cost. A true prompted large 4-way router runs only when API keys are present; offline runs use a cue rubric labelled `live_llm=false`. Single-seed FT point estimates are supplemented by bootstrap and k -fold, but multi-seed grids remain future work. Live FT+RAG generation quality and tool-execution sets (e.g. BFCL) remain future work. Safety-sensitive under-escalation is discussed but not stress-tested on a dedicated safety suite.

Ethical Considerations

Mild safety examples are routed to refusal-capable `answer_small`, never `tools`. Under-escalation of safety-sensitive queries is treated as a first-class risk for any live design. Synthetic workplace data may not represent real employee populations; do not deploy without privacy review of any future production logs. Public remaps inherit licenses/terms of SQuAD, HotpotQA, and CLINC-150; redistribution follows those sources' terms.

5 Conclusion

A 4-way FT-first behavioural router is measured as a classifier and costed as a simulation. FT DistilBERT learns the labelling policy better than strong non-neural baselines in-domain and under a hard OOD probe, with residual risk in tools over-prediction (in-domain) and under-escalation (OOD). Paired McNemar / bootstrap tests on `eval_v2` quantify DistilBERT vs MiniLM vs keyword without inventing significance. FT+RAG is the `rag` path: the small model answers from retrieved documents, so new facts do not require retraining the behavioural weights. Adaptive RAG and cascading are adopted as prior art, not claimed as new.

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. [Self-RAG: Learning to retrieve, generate, and critique through self-reflection](#). *arXiv preprint arXiv:2310.11511*.
- Lingjiao Chen, Matei Zaharia, and James Zou. 2023. [FrugalGPT: How to use large language models while reducing cost and improving performance](#). *arXiv preprint arXiv:2305.05176*.
- Shufan Chen, He Zheng, and Lei Cui. 2025. [When and how to augment your input: Question routing helps balance the accuracy and efficiency of large language models](#). In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 3621–3634, Albuquerque, New Mexico. Association for Computational Linguistics. QueryRouter framework.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. 2024. [Hybrid LLM: Cost-efficient and quality-aware query routing](#). In *The Twelfth International Conference on Learning Representations*. Also arXiv:2404.14618.
- Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C. Park. 2024. [Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7036–7050, Mexico City, Mexico. Association for Computational Linguistics.
- Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. 2019. [An evaluation dataset for intent classification and out-of-scope prediction](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1311–1316, Hong Kong, China. Association for Computational Linguistics.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Advances in Neural Information Processing Systems*.
- Aman Madaan, Pranjal Aggarwal, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. 2024. [AutoMix: Automatically mixing language models](#). In *Advances in Neural Information Processing Systems*. Also arXiv:2310.12963.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. 2025. [RouteLLM: Learning to route LLMs with preference data](#). In *The Thirteenth International Conference on Learning Representations*. Also arXiv:2406.18665.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. [SQuAD: 100,000+ questions for machine comprehension of text](#). In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, Austin, Texas. Association for Computational Linguistics.

- Yuanzhe Shen, Yide Liu, Zisu Huang, Ruicheng Yin, Xiaoping Zheng, and Xuanjing Huang. 2025. [SATER: A self-aware and token-efficient approach to routing and cascading](#). *arXiv preprint arXiv:2510.05164*. EMNLP 2025.
- Jiejun Tan, Zhicheng Dou, Yutao Zhu, Peidong Guo, Kun Fang, and Ji-Rong Wen. 2024. [Small models, big insights: Leveraging slim proxy models to decide when and what to retrieve for LLMs](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4420–4436, Bangkok, Thailand. Association for Computational Linguistics. SlimPLM; also arXiv:2402.12052.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. [CRAG—corrective retrieval augmented generation](#). *arXiv preprint arXiv:2401.15884*.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. [HotpotQA: A dataset for diverse, explainable multi-hop question answering](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium. Association for Computational Linguistics.
- Woongyeong Yeo, Kangsan Kim, Soyeong Jeong, Jinheon Baek, and Sung Ju Hwang. 2025. [UniversalRAG: Retrieval-augmented generation over corpora of diverse modalities and granularities](#). *arXiv preprint arXiv:2504.20734*.
- Tian Yu, Shaolei Zhang, and Yang Feng. 2024. [Auto-RAG: Autonomous retrieval-augmented generation for large language models](#). *arXiv preprint arXiv:2411.19443*.
- Jiarui Zhang, Xiangyu Liu, Yong Hu, Chaoyue Niu, Fan Wu, and Guihai Chen. 2025. [RAGRouter: Learning to route queries to multiple retrieval-augmented language models](#). *arXiv preprint arXiv:2505.23052*.